HAMER GLYCAN SYSTEMS · WHITE PAPER SERIES · REVISED EDITION

Dietary Xenoantigen Load as a Modifiable Amplifier of Chronic Lyme Disease Sequelae

The 'Persistent Fire' Model — impaired lymphatic clearance and microvascular transport as convergent amplifiers of post-treatment Lyme disease syndrome

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> **Firewall convention.** Every substantive claim is labelled ESTABLISHED (published, replicated, peer-reviewed), CONTESTED (published evidence conflicting or disputed), or HYPOTHESIS (a proposition of this framework, not yet tested). No hypothesis is asserted as fact, and no established finding is claimed as original to this work.

Abstract

Background. Post-Treatment Lyme Disease Syndrome (PTLDS) affects an estimated 10–20% of treated Lyme patients, producing fatigue, musculoskeletal pain and neurocognitive dysfunction persisting beyond six months after antibiotic therapy, in the absence of microbiological evidence of ongoing infection. No disease-modifying intervention is approved, and extended antimicrobial courses have not demonstrated durable benefit in controlled trials.

Hypothesis. This paper proposes that dietary xenoantigens — N-glycolylneuraminic acid (Neu5Gc) from bovine dairy and red meat, and wheat amylase-trypsin inhibitors (ATIs) — sustain a low-grade systemic cytokine burden that impairs lymphatic contractile function and elevates interstitial fluid pressure, thereby degrading the clearance of inflammatory debris and residual antigen from tissue. The mechanism is proposed as a modifiable amplifier of symptom persistence, not as a cause of Lyme disease.

Two distinct claims. The revised model separates arguments that the previous edition conflated. In *active or disseminated infection*, the relevant argument is transport: elevated interstitial fluid pressure and impaired perfusion reduce antimicrobial penetration into deep tissue. In *PTLDS*, where bacterial clearance is presumed complete, the relevant argument is drainage: impaired lymphatic pumping retards removal of residual antigen, immune complexes and inflammatory debris. These require different evidence and different trials, and are treated separately below.

1. The Recovery Variability Problem

Lyme disease is the most common vector-borne illness in the Northern Hemisphere, with an estimated 476,000 cases diagnosed and treated annually in the United States. Most patients recover fully following a standard two-to-four week antibiotic course. A clinically significant minority do not.

The pathophysiology of PTLDS remains contested. Proposed mechanisms include residual Borrelia antigen persistence, molecular mimicry between Borrelia epitopes and host tissue antigens,

antibiotic-induced microbiome dysbiosis, and persistent immune dysregulation independent of bacterial presence. None has achieved consensus.

The question this paper addresses is narrower than PTLDS causation: why do patients with apparently equivalent infections and equivalent treatment diverge so sharply in outcome? Divergence of that kind implies modifiable secondary factors. This paper proposes that dietary xenoantigen load is one such factor, acting not on the pathogen but on the tissue transport systems responsible for resolving its aftermath.

2. ESTABLISHED — Inflammation Suppresses Lymphatic Function

This section carries the weight of the paper, and none of it is original to this framework. The relationship between circulating inflammatory cytokines and lymphatic contractile failure is established physiology, demonstrated across multiple independent models.

2.1 Cytokines suppress lymphatic pumping, and the effect is systemic

Using near-infrared fluorescence imaging to observe lymphatic propulsion non-invasively, investigators injected LPS into the dorsal hind paw of mice and observed severe retardation of lymphatic pumping, with cessation of propulsion detectable as early as four hours. The critical finding was that the effect was not confined to the injected limb: propulsion ceased bilaterally following a unilateral insult. Separate intradermal administration of IL-6, TNF-alpha and IL-1beta — each elevated in serum during the pumping cessation — independently reduced systemic lymphatic pulsing frequency and velocity.

The systemic character of this response is what makes dietary input mechanistically relevant. An inflammatory stimulus arising at the intestinal barrier need not act locally on the tissue of interest. Circulating cytokines suppress lymphatic propulsion at anatomically distant sites.

2.2 The mechanism is nitric-oxide mediated, and it is reversible

The suppressive effect of IL-1beta on lymphatic pumping was abated by pretreatment with an inhibitor of inducible nitric oxide synthase. In TNF-transgenic mice with inflammatory arthritis, lymphatic drainage declines through loss of intrinsic lymphatic contractions, and pharmacologic iNOS inhibition has been shown to recover popliteal lymphatic vessel contractions and restore clearance on near-infrared imaging.

Reversibility is the point on which this entire model depends. Lymphatic contractile failure under inflammatory load is not structural damage. It is a functional state that resolves when the inflammatory signal is withdrawn. If a dietary input is sustaining that signal, withdrawing the input is a plausible route to restoring drainage — and the prediction is specific enough to fail.

2.3 Cytokines also increase lymphatic endothelial permeability

Beyond contractility, lymphatic vessel permeability is increased by TNF-alpha, IL-1beta, histamine, and the VEGF-C/VEGFR-3 axis, while lymphatic contraction is negatively regulated by

prostaglandins, histamine and nitric oxide. A leaking, non-contracting collecting vessel does not merely fail to clear the interstitium; it returns fluid to it.

3. ESTABLISHED — Interstitial Pressure as a Transport Barrier

The second established body of work comes from oncology, where the problem of delivering drugs into poorly drained tissue has been studied for four decades.

Elevated interstitial fluid pressure is recognised as a hallmark of solid tumours and as a direct compromise to the delivery of therapeutics. Work from the Steele Laboratory demonstrated that normalising the vasculature reduces interstitial fluid pressure and induces a hydrostatic pressure gradient across the vessel wall, producing deeper penetration of molecules into tissue. The framework has since been extended and modelled across agent sizes and delivery modalities.

The relevance here is structural rather than oncological. Where interstitial pressure is high and lymphatic drainage is impaired, the transvascular pressure gradient that drives drug into tissue is lost. This is a general transport problem, not a property of tumours, and it applies wherever a pathogen occupies a poorly drained compartment.

4. ESTABLISHED — Borrelia Occupies the Compartments This Affects

- Borrelia preferentially colonises lymph nodes, synovial tissue and the central nervous system — compartments characterised by slow fluid flow, restricted immune effector access and limited antibiotic penetration.
- It modifies the local immune environment, downregulating complement activation at its surface and suppressing macrophage bactericidal activity.

A terminological distinction worth making explicitly. Borrelia acts through molecular mimicry — a foreign epitope structurally resembling a host antigen. Neu5Gc does not. Neu5Gc is a xenoautoantigen: a genuinely foreign sugar metabolically incorporated into host tissue and subsequently recognised as foreign. These are different mechanisms with different immunological consequences, and the distinction should not be blurred simply because both produce responses against host tissue.

Not established. Three further properties are often attributed to Borrelia and bear directly on this model, but they do not carry the weight of the points above.

CONTESTED — biofilm. Borrelia forms biofilm-like aggregates that resist antibiotics in culture, and such aggregates have been reported in some tissue specimens. Their clinical significance in human infection is disputed. The companion paper in this series, Glycan Determinants of Biofilm Persistence, takes up this question from a different angle.

CONTESTED — molecular mimicry. Post-infectious autoimmunity is well supported in antibiotic-refractory Lyme arthritis, but the best-known proposed mimic, between outer surface protein A and human LFA-1, did not hold up on later study. That cross-reactive responses persist after bacterial clearance is plausible; the mechanism is not settled.

HYPOTHESIS — interstitial stasis. That Borrelia exploits interstitial stasis — that reduced lymphatic flow and raised interstitial viscosity create protected niches with poor immune surveillance and drug delivery — is a proposition of this paper, not a published finding. It is the bridge between the established physiology of Sections 2 and 3 and the model in Section 5.

5. HYPOTHESIS — The Persistent Fire Model

5.1 The proposition

Chronic dietary exposure to Neu5Gc and wheat ATIs sustains a low-grade systemic elevation of TNF-alpha, IL-6 and IL-1beta. By the mechanisms established in Section 2, this elevation suppresses lymphatic contractile function and increases lymphatic and vascular endothelial permeability. By the mechanisms established in Section 3, the resulting interstitial accumulation raises tissue pressure and degrades transvascular transport. By the biology established in Section 4, these are precisely the conditions Borrelia is adapted to exploit and precisely the compartments in which post-infectious debris must be cleared.

What is hypothesised is the magnitude of the input, not the chain it acts on. The cited studies use LPS injection and TNF-transgenic models — inflammatory loads far exceeding anything produced by ordinary diet. Whether habitual dietary xenoantigen exposure generates cytokine concentrations sufficient to measurably suppress lymphatic pumping in humans is unmeasured, and is the single weakest quantitative link in this argument.

5.2 Two claims, separated

	Active / disseminated Lyme	PTLDS
Bacterial state	Present, in sanctuary sites	Presumed cleared
Relevant argument	Transport — antimicrobial delivery	Drainage — debris and antigen clearance
Supporting framework	Interstitial pressure and vascular normalisation (Section 3)	Cytokine suppression of lymphatic pumping (Section 2)
Predicted benefit of elimination	Improved drug penetration into poorly perfused tissue	Improved clearance of residual antigen and immune complexes
Principal caution	Inflammation also mediates bacterial killing; suppression is not unambiguously beneficial during active infection	Weaker — with the pathogen cleared, inflammatory tone has less protective role

The previous edition of this paper argued both positions simultaneously. They are separated here because they require different trial designs, carry different risks, and would be judged against different literatures. The PTLDS claim is the stronger of the two and should lead.

5.3 Why PTLDS patients specifically

This model offers an answer that the previous edition missed, and it is arguably its most useful contribution. Every PTLDS patient has, by definition, completed two to four weeks of antibiotic therapy. Antibiotic-induced dysbiosis and intestinal barrier disruption are already listed among the leading competing explanations for PTLDS. Under the present model they are not a competing explanation at all — they are the priming event. Treatment itself opens the barrier through which dietary xenoantigen reaches the systemic circulation, at precisely the moment the patient most needs lymphatic clearance to be working.

This predicts that dietary xenoantigen load should matter more in the post-treatment period than at any other point in the disease course, and that the subset of patients with the greatest antibiotic-associated barrier disruption should show the strongest association between dietary burden and symptom persistence.

5.4 The analogy

The Borrelia infection is the fire. Antibiotic therapy extinguishes the visible flames but leaves embers — residual antigen, dysregulated immune response, and sanctuary sites the drug could not reach. Dietary xenoantigens are the bellows: each exposure supplies a fresh inflammatory input that sustains the vascular permeability, lymphatic suppression and cytokine elevation on which the embers depend. Withdraw the bellows and the embers are not immediately extinguished — cross-reactive autoimmunity, microbiome disruption and direct tissue effects each have their own resolution timelines — but they lose their amplification.

6. HYPOTHESIS — Neurocognitive Symptoms

Cognitive impairment is among the most debilitating features of PTLDS and has been attributed to neuroinflammation, cerebrovascular dysfunction, microglial activation and direct Borrelia neurotropism.

The dietary xenoantigen framework adds a contribution through two routes: cerebrovascular endothelial activation by the mechanisms described above, and systemic cytokine elevation reaching the central nervous system through saturable transport mechanisms and through circumventricular organs lacking a functional blood-brain barrier. ATI-driven TLR4 activation contributes independently through the same channels. Under this model, a patient consuming a typical Western diet would carry a neuroinflammatory baseline additional to any Borrelia-associated component. That baseline has not been measured.

The prediction is specific and cheap to test: neurocognitive improvement — processing speed, working memory, attention — should become measurable within a 7–10 day window following dietary elimination, independent of any change in bacterial burden. Daily assessment is achievable with validated computerised batteries and requires no laboratory infrastructure.

7. Predictions, and a Design Problem That Must Be Solved First

The stratification problem. An elimination trial in unselected PTLDS patients is predicted by this framework to fail. Anti-Neu5Gc antibody repertoires are polyclonal, epitope-specific and variable between individuals; recognition depends on the extended glycan epitope rather than on Neu5Gc content. Only the subset of patients whose repertoires recognise the specific glycotopes present in bovine dietary sources would be expected to respond. Recruiting without stratification dilutes any true effect toward null, and a null result would then be reported as refuting the hypothesis when it had in fact refuted the sampling. Every trial below must stratify on antibody profile before randomisation.

Prediction	Design	Expected finding
P1 — Epitope profile, not titre, predicts symptom burden	Sialoglycan microarray profiling of PTLDS patients vs recovered Lyme patients vs controls; intestinal permeability markers (FABP2, zonulin); standardised symptom severity scale; dietary burden by FFQ	Bovine-reactive epitope profiles over-represented in PTLDS; total anti-Neu5Gc titre alone shows weak or no association
P2 — Elimination reduces symptom burden in stratified patients	Elimination trial in confirmed PTLDS, stratified by epitope profile; symptom scale at 0, 4, 8, 12 weeks; hsCRP, TNF-alpha, IL-6, FABP2, neurocognitive battery	Symptom reduction confined to, or substantially greater in, the bovine-reactive stratum
P3 — Neurocognitive gain within the clearance window	Daily computerised neurocognitive assessment during 14-day elimination; processing speed, working memory, attention at days 0, 7, 10, 14	Measurable improvement beginning day 7–10, inconsistent with an antimicrobial mechanism
P4 — Lymphatic transit is measurably impaired and improves on elimination	Near-infrared fluorescence lymphatic imaging before and after 12-week elimination; lymphatic transit velocity, packet frequency, interstitial fluid volume	Reduced baseline transit velocity in PTLDS; measurable recovery post-elimination. This is the direct test of Section 2 and the most decisive prediction in the paper
P5 — Combined intervention outperforms supportive care	RCT: supportive care vs supportive care plus stratified dietary elimination; symptom scale at 12 weeks; inflammatory markers, lymphatic imaging	Superior symptom reduction in combined arm

P4 is the prediction to run first. It measures the proposed mechanism directly rather than inferring it from symptoms, it uses an established imaging modality, and it does not depend on the epitope stratification question being settled beforehand. If lymphatic transit velocity in PTLDS patients is normal, the central mechanism of this paper is wrong and the remaining predictions need not be tested.

8. On Repeated Antimicrobial Courses

Controlled trials of extended antibiotic therapy in PTLDS have not demonstrated durable benefit, and prolonged exposure carries non-trivial risk: antibiotic-associated colitis, further microbiome disruption, selection for resistant organisms, and catheter-related complications from intravenous administration.

The present model offers one mechanistic account of why the approach underperforms. If a component of the persistent symptom burden arises from impaired tissue clearance rather than from surviving bacteria, an antimicrobial addresses a target that is no longer the limiting factor. Dietary elimination is not proposed as a replacement for antimicrobial therapy, and this paper takes no position on the duration or choice of antibiotic regimen. It proposes an adjunct addressing a different variable.

9. Limitations

- No clinical data on dietary xenoantigen exposure in PTLDS patients have been collected, and no elimination trial in this population has been conducted. This is a hypothesis paper.
- The magnitude question is unresolved. Every cited demonstration of cytokine-mediated lymphatic suppression uses inflammatory loads far above those plausibly produced by diet. That habitual dietary exposure reaches the necessary threshold is assumed, not shown.
- Lymphatic transit velocity has never been measured in PTLDS patients, stratified by diet or otherwise. Section 2 establishes that the mechanism exists; it does not establish that it operates here.
- The relative contribution of dietary neuroinflammation against Borrelia-associated neuroinflammation has not been quantified, and the two are not separable by current methods in living patients.
- Reducing inflammatory tone is not unambiguously beneficial during active infection, where inflammation also mediates bacterial killing. The model is on firmer ground in the post-treatment setting than during treatment.
- PTLDS is contested clinical territory. This paper does not adjudicate the persistence debate, does not claim to account for all PTLDS, and is not offered as an alternative to the differential diagnosis that any persistent symptom burden warrants.

10. Conclusion

PTLDS describes patients whose pathogen has been treated and whose symptoms remain. Existing frameworks concentrate on the biology of Borrelia and the pharmacology of antibiotics. This paper proposes examining a third variable: the tissue transport systems responsible for clearing what infection and treatment leave behind, and the possibility that those systems are being held in a suppressed state by an input nobody has measured.

The mechanism by which circulating cytokines suppress lymphatic contractility is established, and its reversibility on removal of the inflammatory signal is demonstrated. What is proposed here is that

habitual diet may constitute one such signal. That proposition is testable by direct imaging in a modest cohort, and it should be tested before it is believed.

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